

Replication: Review Solicitation and Future Spending

Randomized encouragement, ITT, LATE, heterogeneity, and multiple testing in Karaman (2026)

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1 Research Question

- **Research question.**
 - Does posting an online review change the reviewer’s future spending on the reviewed brand, and
 - does soliciting a review change spending by inducing customers to post?
 - The paper also asks whether these effects differ by customer satisfaction and loyalty.
- **Mechanism.** The causal story is that solicitation increases the probability of posting a review, and posting a review makes the customer’s evaluation of the stay more salient and cognitively accessible. For dissatisfied customers, writing the review reinforces a negative experience, so future spending on the reviewed brand falls. For highly loyal customers, even a less-than-perfect positive review can make disappointment salient enough to reduce future spending. For perfectly satisfied customers, posting a positive review does not meaningfully change future spending.

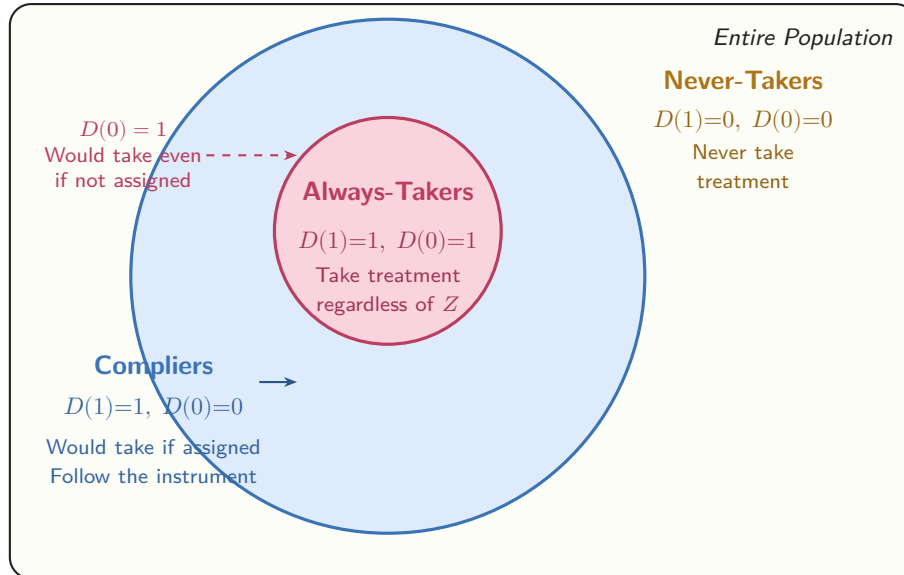
Why this design. Posting is self-selected, so randomized review solicitation is appropriate as an encouragement/IV for actual posting, while also identifying the ITT effect of solicitation itself.

To fully understand the LATE estimator, let’s consider the following plot:

LATE vs. ATE: Who Does Each Estimator Cover?

(A) Population Subgroups under an Instrument Z

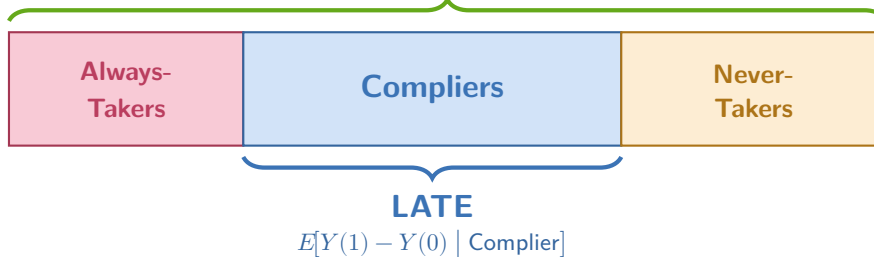
Assuming monotonicity (no defiers), individuals fall into three types based on potential treatment $D(z)$:



(B) What Each Estimator Averages Over

$E[Y(1) - Y(0)]$ — averaged over *everyone*

ATE



Key insight:

IV/2SLS estimates LATE, not ATE. It captures the causal effect *only* for compliers—those whose treatment is actually changed by the instrument. If compliers differ from always-/never-takers, $\text{LATE} \neq \text{ATE}$.

And in this experiment, the design is:

2 Data, timing, and satisfaction-to-valence mapping

This section defines the timing, variables, and subgroup cells needed to interpret the first-stage, ITT, and LATE estimates that follow.

The unit of observation is an individual loyalty-program customer who completed a June 2014 post-stay satisfaction survey. The treatment assignment is review solicitation, randomized after survey completion.

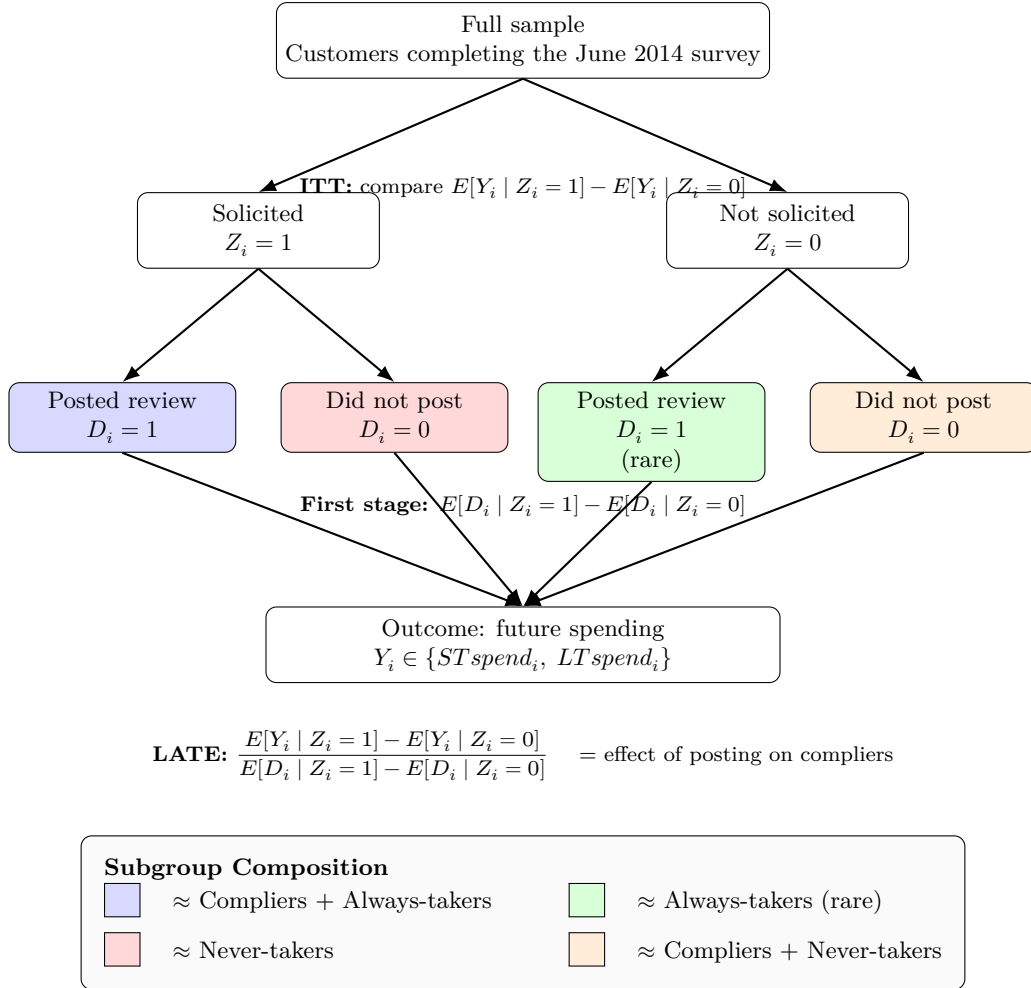
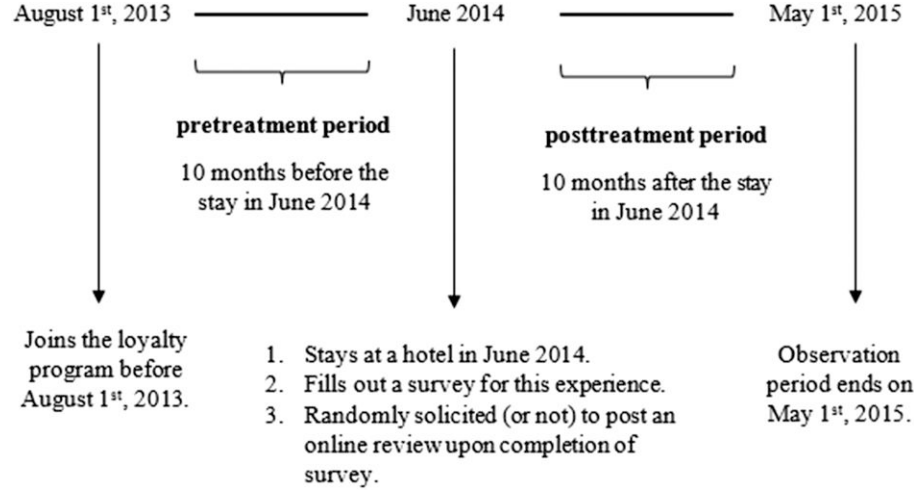


Figure 1: Encouragement design underlying the ITT and LATE estimates.

The endogenous treatment is actual review posting, which occurs only after solicitation or non-solicitation status is realized.



- Pre-period: spending history before the June 2014 stay and survey.
- Assignment point: customer completes the June 2014 survey and is randomly solicited or not solicited to post a review.
- Post-period: future spending is measured after assignment.
- Short-term outcome: **STspend_i**, spending in the first post-treatment window.
- Long-term outcome: **LTspend_i**, spending in the later post-treatment window.

Core variables.

- **Solicit_i**: indicator equal to 1 if customer *i* was randomly solicited to post a review.
- **Review_i**: indicator equal to 1 if customer *i* actually posted a review.
- **STspend_i**: short-term future spending.
- **LTspend_i**: long-term future spending.
- **Satisfaction_i**: survey satisfaction score from 1 to 10.
- **Loyalty_i**: loyalty tier: Silver, Gold, or Platinum.

Pre-treatment versus post-treatment variables.

- Pre-treatment or assignment-time variables: satisfaction score, loyalty tier, solicitation assignment.
- Post-treatment variables: actual review posting and future spending.
- This ordering matters because heterogeneity by satisfaction and loyalty does not condition on outcomes or post-treatment behavior.

Figure 2 shows that satisfaction scores 1–6 correspond to negative reviews, while scores 8–10 correspond to positive reviews. This justifies the main satisfaction grouping: 6, 8, 9, and 10.

Subgroup construction. The heterogeneous estimates use 12 mutually exclusive loyalty-by-satisfaction cells:

- Silver: Silver6, Silver8, Silver9, Silver10
- Gold: Gold6, Gold8, Gold9, Gold10
- Platinum: Platinum6, Platinum8, Platinum9, Platinum10

The ≤ 6 cells pool highly dissatisfied customers because their reviews are negative. Scores 8, 9, and 10 are kept separate because the paper's core asymmetry depends on distinguishing perfect satisfaction from less-than-perfect positive experiences.

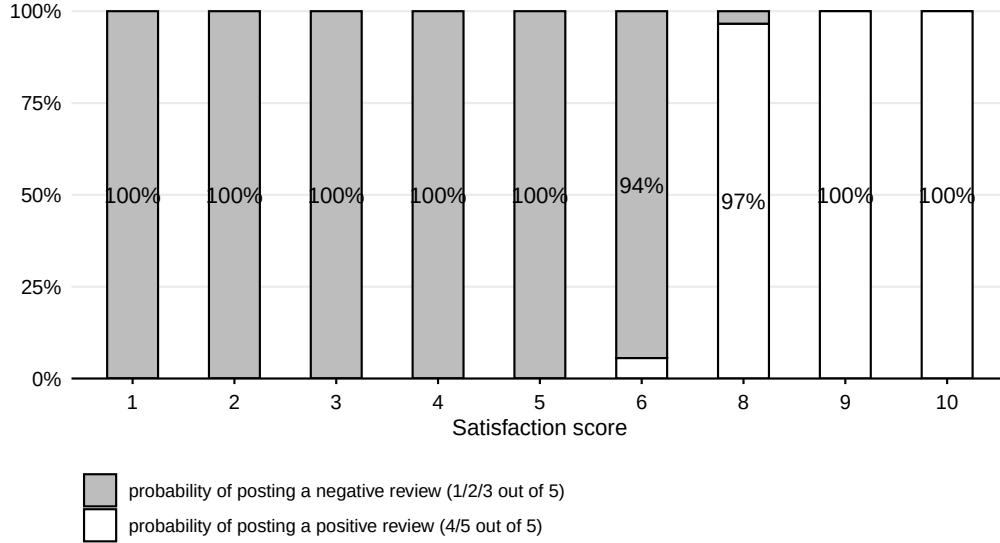


Figure 2: Probability of posting a negative versus positive review by satisfaction score

3 First stage — solicitation drives review posting

Argument: Random solicitation strongly increases actual review posting, establishing instrument relevance and motivating the one-sided-noncompliance interpretation used in the LATE section.

Evidence 1: Solicited customers are much more likely to post reviews, while unsolicited customers almost never post. Figure 3 visualizes the probability of observing a posted review across solicitation status, satisfaction groups, and loyalty tiers. The pattern shows a clear first stage: solicitation substantially increases the probability of review posting in every subgroup.

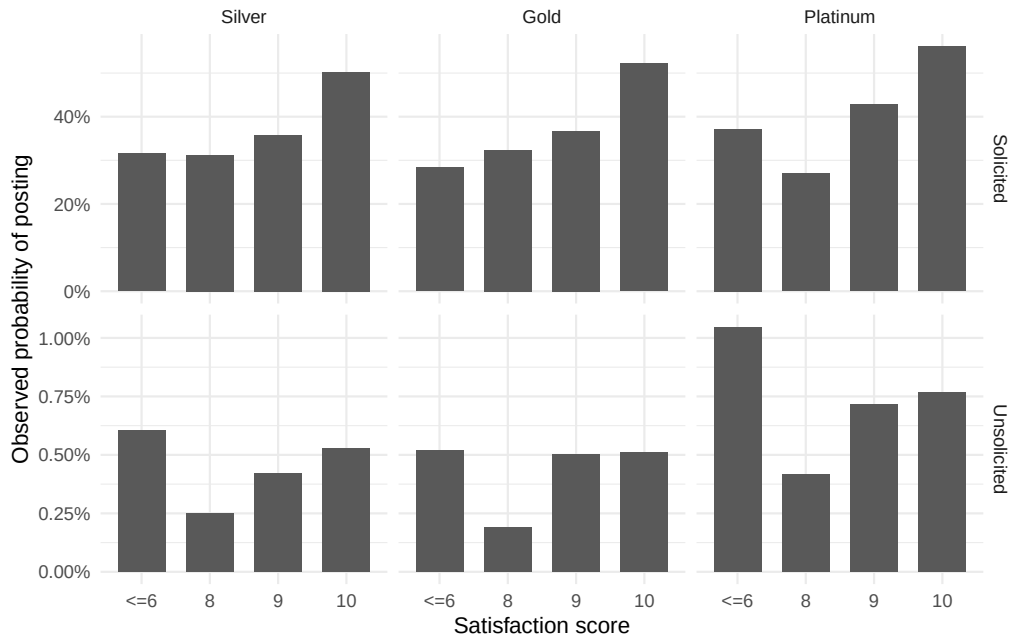


Figure 3: One-sided noncompliance: review posting by solicitation status

Evidence 2: In Online Appendix C, Table OC1, the author estimates first-stage regressions overall and separately within the 12 loyalty-by-satisfaction cells. These regressions test the IV relevance condition:

solicitation must strongly predict actual review posting.

$$Review_i = \pi_0 + \pi_1 Solicit_i + v_i$$

Table 1: First-stage estimates for review posting

Tier	Score	Solicit coefficient (SE)	Robust F	p-value	N
All customers		0.39 (0.01)	705.13	0.000	10000
Silver	<=6	0.31 (0.04)	50.32	0.000	941
	8	0.31 (0.04)	71.35	0.000	1360
	9	0.35 (0.04)	95.51	0.000	1608
	10	0.50 (0.03)	209.81	0.000	2104
Gold	<=6	0.28 (0.06)	20.00	0.000	437
	8	0.32 (0.06)	32.00	0.000	597
	9	0.36 (0.06)	38.25	0.000	666
	10	0.52 (0.06)	71.56	0.000	848
Platinum	<=6	0.36 (0.07)	23.63	0.000	234
	8	0.27 (0.07)	13.15	0.000	277
	9	0.42 (0.06)	50.10	0.000	349
	10	0.55 (0.07)	70.50	0.000	579

4 ITT — solicitation changes future spending

Argument: Randomly assigning customers to review solicitation reduces future spending on the solicited brand, with the strongest effects among dissatisfied customers and less-than-perfect Platinum customers.

To establish the argument, the section first uses the overall ITT regression to test whether randomized solicitation reduces future spending on average. It then uses subgroup ITT estimates to show that this average effect is driven by dissatisfied customers and less-than-perfect Platinum customers. The score-10 estimates serve as the contrast case, showing that solicitation is not uniformly harmful.

Evidence 1: Overall ITT

- **Claim:** Solicitation reduces future spending on the solicited brand on average.
- **Table/Figure:** \label{tab:itt_overall} — Table 2
- **Specification:** The paper estimates:

$$FutureSpending_i^h = \gamma_0 + \gamma_1 Solicitation_i + \varepsilon_i,$$

where h indexes the short-term and long-term spending windows.

- **Key coefficient:** γ_1 , the coefficient on $Solicitation_i$.
 - **Direction:** The solicitation coefficient is negative for both short-term and long-term spending.
 - **Interpretation:** Random assignment to solicitation lowers future spending on the solicited brand.
 - **Design role:** This is the reduced-form effect needed before estimating the LATE of actual review posting.

Table 2: ITT Effect on Future Spending on the Solicited Brand

Outcome	ITT	N
Short-term future spending	-22.91 (15.17)	0
Long-term future spending	-23.19** (11.24)	0

Evidence 2: Heterogeneous ITT by satisfaction and loyalty

- **Claim:** The negative ITT effect is concentrated among highly dissatisfied customers and less-than-perfect Platinum customers.
- **Table/Figure:** \label{tab:itt_hetero} — Table 3
- **Specification:** The paper estimates the ITT separately within each loyalty-by-satisfaction cell g :

$$FutureSpending_i^h = \gamma_{0g} + \gamma_{1g}Solicitation_i + \varepsilon_i,$$

where h indexes the short-term and long-term spending windows.

- **Key coefficient:** γ_{1g} , the coefficient on $Solicitation_i$.
 - **Highly dissatisfied customers:** In the ≤ 6 satisfaction column, the solicitation coefficients are negative across all loyalty tiers and both time horizons. In the replication table, the largest magnitudes appear for Platinum customers: -78.96 in the short term and -84.78 in the long term.
 - **Less-than-perfect Platinum customers:** Within the Platinum panel, solicitation also reduces spending for customers with satisfaction scores 8 and 9. The score-8 short-term estimate is -95.96 , and the score-9 long-term estimate is -96.40 .
 - **Perfectly satisfied customers as contrast:** For score-10 customers, the estimates are not statistically significant across loyalty tiers. This shows that solicitation is not mechanically harmful for everyone.
 - **Design interpretation:** The ≤ 6 cells correspond to negative reviews from Figure 2, while Platinum scores 8 and 9 correspond to positive but less-than-perfect experiences. Thus, the heterogeneous ITT pattern supports the paper’s asymmetry: solicitation is most harmful when it induces reviews from dissatisfied customers, and among the most loyal customers, even less-than-perfect positive experiences can reduce future spending.
 - **Robustness link:** Because Table ?? tests many subgroup effects, this pattern should be read together with the Romano–Wolf correction in Table 6.

Table 3: ITT Effects on Future Spending on the Solicited Brand

Horizon	term	Satisfaction			
		<=6	8	9	10
Silver					
Short-term (0-5 months)	Solicitation	-10.998* (6.542)	-6.679 (6.692)	-3.059 (35.867)	-6.343 (30.551)
Long-term (6-10 months)	Solicitation	-8.604* (4.501)	3.470 (5.745)	-4.475 (21.732)	-6.956 (15.092)
	No. of observations	941	1,360	1,608	2,104
Gold					
Short-term (0-5 months)	Solicitation	-58.566* (31.701)	-42.736 (41.291)	-10.388 (40.144)	-25.006 (130.057)
Long-term (6-10 months)	Solicitation	-39.446 (42.169)	-25.012 (48.060)	-9.367 (43.466)	-33.766 (77.409)
	No. of observations	437	597	666	848
Platinum					
Short-term (0-5 months)	Solicitation	-78.958* (46.443)	-95.959* (56.512)	-69.491 (45.197)	-44.427 (136.862)
Long-term (6-10 months)	Solicitation	-84.782* (50.327)	-92.201 (85.581)	-96.400* (49.914)	-36.076 (98.244)
	No. of observations	234	277	349	579

Notes: All estimations include a constant term that is not tabulated for confidentiality reasons. Robust standard errors are shown in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

5 LATE — posting changes future spending for compliers

Argument: Instrumenting actual review posting with randomized solicitation shows that posting a review reduces future spending, especially for dissatisfied reviewers and less-than-perfect Platinum reviewers.

To establish this argument, the section first estimates the overall IV/LATE effect by instrumenting (*Review_i*) with (*Solicitation_i*). This converts the reduced-form ITT from Section 4 into the causal effect of actual review posting for compliers. The section then estimates the same IV specification within the 12 loyalty-by-satisfaction cells to show that the largest effects occur in the same cells where the ITT effects were strongest.

Evidence 1: Overall LATE

- **Claim:** Posting a review reduces future spending on the reviewed brand on average.
- **Table/Figure:** \label{tab:late_overall} — Table 4
- **Specification:** The paper estimates:

$$FutureSpending_i^h = \alpha_0 + \tau Review_i + u_i,$$

with *Review_i* instrumented by *Solicitation_i*, where *h* indexes the short-term and long-term spending windows.

- **Key coefficient:** τ , the IV coefficient on *Review_i*.
 - **Direction:** The review coefficient is negative for both short-term and long-term spending.
 - **Magnitude:** Posting reduces future spending by roughly the ITT effect divided by the first-stage effect.

- **Interpretation:** Among customers whose posting behavior is shifted by solicitation, posting a review lowers future spending on the reviewed brand.
- **Design role:** This is the causal counterpart to the ITT estimate in Section 4: ITT measures assignment to solicitation, while LATE measures actual posting induced by solicitation.
- **Robustness link:** This estimate depends on the first stage in Section 3 and the IV assumptions from Phase A.

Table 4: Overall LATE of Posting a Review on Future Spending

	Short-term (0-5 months)	Long-term (6-10 months)
Review	−59.236 (39.383)	−59.945** (29.139)
No. of observations	10.000	10.000

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: All estimations include a constant term that is not tabulated for confidentiality reasons. Robust standard errors are shown in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Evidence 2: Heterogeneous LATE by satisfaction and loyalty

- **Claim:** The negative posting effect is concentrated among highly dissatisfied reviewers and less-than-perfect Platinum reviewers.
- **Table/Figure:** \label{tab:late_hetero} — Table 5
- **Specification:** The paper estimates the IV model separately within each loyalty-by-satisfaction cell g :

$$FutureSpending_i^h = \alpha_{0g} + \tau_g Review_i + u_i,$$

with $Review_i$ instrumented by $Solicitation_i$.

- **Key coefficient:** τ_g , the IV coefficient on $Review_i$.
- **Highly dissatisfied reviewers:** In the ≤ 6 satisfaction column, the LATE estimates are negative across loyalty tiers. In the published long-term results, posting reduces spending by about \$50 for Silver, \$122 for Gold, and \$211 for Platinum reviewers.
- **Less-than-perfect Platinum reviewers:** In the Platinum panel, posting reduces long-term spending for score-8 and score-9 customers. The published estimates are about −\$250 for score 8 and −\$191 for score 9.
- **Perfectly satisfied reviewers as contrast:** For score-10 customers, the estimates are not statistically significant across loyalty tiers. This shows that posting is not mechanically harmful for everyone.
- **Design interpretation:** The ≤ 6 cells correspond to negative reviews, while Platinum score-8 and score-9 cells correspond to positive but less-than-perfect experiences. Thus, the LATE results sharpen the Section 4 ITT pattern: the spending decline operates through induced review posting.
- **Robustness link:** Because Table 5 tests many subgroup effects, these estimates should be read together with the Romano–Wolf correction in Table 6.

Table 5: LATE Effects of Posting a Review on Future Spending on the Reviewed Brand

		Satisfaction			
Horizon	term	<=6	8	9	10
Silver					
Short-term (0-5 months)	Review	-35.505* (21.518)	-21.544 (21.496)	-8.646 (101.371)	-12.760 (61.557)
Long-term (6-10 months)	Review	-27.777* (15.186)	11.195 (18.573)	-12.651 (61.482)	-13.994 (30.398)
	No. of observations	941	1,360	1,608	2,104
Gold					
Short-term (0-5 months)	Review	-210.813* (116.863)	-132.869 (129.701)	-28.646 (110.361)	-48.342 (251.904)
Long-term (6-10 months)	Review	-141.990 (152.859)	-77.765 (150.217)	-25.831 (119.984)	-65.278 (150.068)
	No. of observations	437	597	666	848
Platinum					
Short-term (0-5 months)	Review	-218.345 (137.178)	-360.609 (239.946)	-164.904 (110.580)	-80.231 (248.111)
Long-term (6-10 months)	Review	-234.448* (130.006)	-346.484 (311.281)	-228.759* (121.343)	-65.149 (176.898)
	No. of observations	234	277	349	579

Notes: All estimations include a constant term that is not tabulated for confidentiality reasons. Robust standard errors are shown in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

6 Robustness — multiple testing and design sensitivity

Argument: The main ITT and LATE patterns remain credible after checking whether they are driven by multiple hypothesis testing, prior reviewer selection, subgroup-specification choices, distributional outliers, or spending substitution across brands.

To assess whether the heterogeneous ITT and LATE patterns are fragile, the available replication table focuses on the Romano–Wolf multiple-testing correction in Table 6. Because the paper estimates many loyalty-by-satisfaction effects, raw subgroup significance can overstate the strength of the evidence; the adjusted p-values ask whether the key cells remain important after controlling family-wise error. The main pattern remains concentrated in the same places as Sections 4 and 5: dissatisfied customers and less-than-perfect Platinum customers. The remaining robustness checks are not reproduced with the public sample data, but each addresses a specific threat in the original paper: Online Appendix F excludes prior review posters to rule out habitual-reviewer selection; Online Appendix E uses a pooled interaction specification to show that the 12-cell pattern is not an artifact of separate regressions; Online Appendix G estimates quantile treatment effects to check that the Platinum dissatisfied result is not driven by outliers; Online Appendix I uses nonreviewed-brand spending to test whether customers simply substitute across company brands; and Online Appendix J discusses survey-participation selection, which limits external validity because satisfaction is observed only for survey takers.

Table 6: Adjusted p-values Under Multiple-Hypothesis Testing

Horizon	term	Satisfaction			
		<=6	8	9	10
Silver					
Short-term (0-5 months)	Raw p-value	(0.093)	(0.318)	(0.932)	(0.836)
	Romano-Wolf p-value	(0.936)	(0.998)	(1.000)	(1.000)
Long-term (6-10 months)	Raw p-value	(0.056)	(0.546)	(0.837)	(0.645)
	Romano-Wolf p-value	(0.879)	(1.000)	(1.000)	(1.000)
Gold					
Short-term (0-5 months)	Raw p-value	(0.065)	(0.301)	(0.796)	(0.848)
	Romano-Wolf p-value	(0.894)	(0.998)	(1.000)	(1.000)
Long-term (6-10 months)	Raw p-value	(0.350)	(0.603)	(0.829)	(0.663)
	Romano-Wolf p-value	(0.998)	(1.000)	(1.000)	(1.000)
Platinum					
Short-term (0-5 months)	Raw p-value	(0.090)	(0.091)	(0.125)	(0.746)
	Romano-Wolf p-value	(0.936)	(0.936)	(0.940)	(1.000)
Long-term (6-10 months)	Raw p-value	(0.093)	(0.282)	(0.054)	(0.714)
	Romano-Wolf p-value	(0.936)	(0.997)	(0.879)	(1.000)

Notes: Raw p-values are based on robust standard errors. Romano-Wolf adjusted p-values are computed with 5000 bootstrap resamples and seed 12345. By design, these adjusted p-values apply to the heterogeneous ITT tests and to the corresponding LATE tests.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

7 Managerial response and design lessons

Purpose: This section translates the ITT, LATE, and robustness results into the design lessons a researcher should take from the paper.

The empirical argument leads to two decision rules. First, solicitation should be targeted: extremely satisfied customers are relatively safe to solicit, while highly dissatisfied customers should not be solicited because solicitation induces reviews that reduce their own future spending. Second, response strategy should prioritize the cells where posting is most harmful: highly dissatisfied reviewers, especially Platinum reviewers, and Platinum reviewers with positive but less-than-perfect experiences.

Figure \label{fig:response_rate} — Paper Figure 5; replication-package Figure 6 — compares the company’s actual response rates across the same loyalty-by-satisfaction cells used in the ITT and LATE tables. The company responds to nearly all highly dissatisfied reviewers, which aligns with the estimated harm in the (6) cells. However, response rates are similar for satisfaction scores 8 and 9 across loyalty tiers, which misses the key heterogeneous result: Platinum customers with less-than-perfect positive experiences are also high-risk.

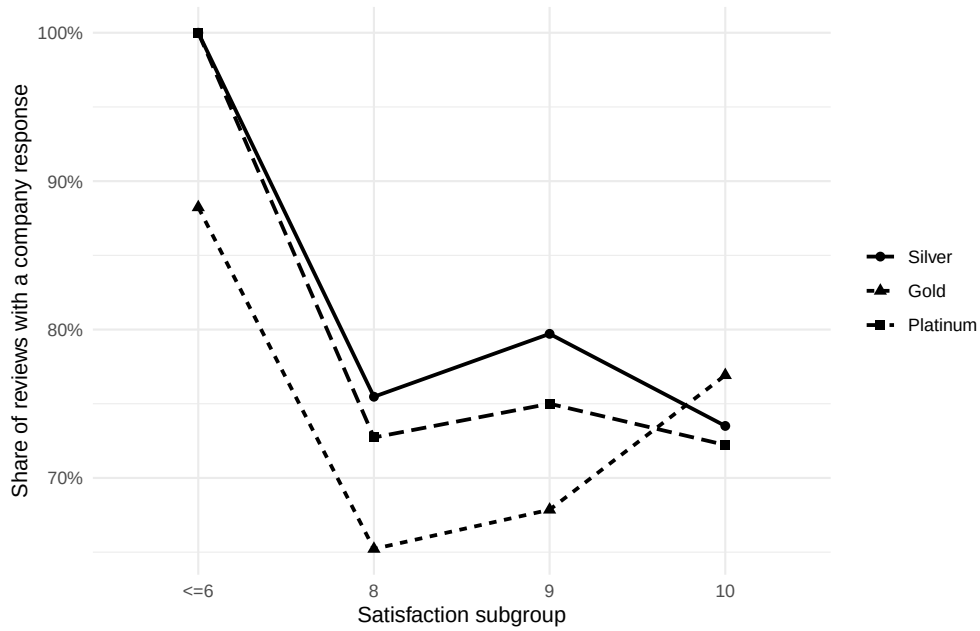
The design lesson is that a randomized encouragement paper should not stop at the average ITT and LATE. The paper builds a full empirical case by linking the first stage, reduced form, IV estimate, heterogeneity cells, multiple-testing correction, and a final descriptive diagnostic on managerial behavior.

Forward pointer: The note’s takeaway is that the same subgroup structure should organize the causal estimates, robustness checks, and practical implications.

Table 7: Management Response Rates by Loyalty and Satisfaction

Loyalty	Satisfaction	No. of reviews	Response rate
Silver	<=6	41	100.0%
Silver	8	53	75.5%
Silver	9	69	79.7%
Silver	10	117	73.5%
Gold	<=6	17	88.2%
Gold	8	23	65.2%
Gold	9	28	67.9%
Gold	10	39	76.9%
Platinum	<=6	18	100.0%
Platinum	8	11	72.7%
Platinum	9	32	75.0%
Platinum	10	36	72.2%

Figure 4: Management response rate by loyalty and satisfaction



8 Appendix

```
#-----
# Figure 3
#-----
valence_data <- review_data %>%
  filter(
    review == 1,
    satisfactionscore %in% c(1, 2, 3, 4, 5, 6, 8, 9, 10)
  ) %>%
  group_by(satisfactionscore) %>%
```

```

summarise(
  `Negative review` = mean(negativerreview, na.rm = TRUE),
  `Positive review` = mean(positivereview, na.rm = TRUE),
  n_reviewers = n(),
  .groups = "drop"
) %>%
pivot_longer(
  cols = c(`Negative review`, `Positive review`),
  names_to = "review_type",
  values_to = "probability"
)
ggplot(valence_data, aes(x = factor(satisfactionscore), y = probability, fill = review_type)) +
  # Create the stacked bars with a black outline
  geom_col(color = "black", width = 0.5) +

  # Add percentage labels only for segments > 10% to prevent clutter
  geom_text(
    aes(label = ifelse(probability > 0.10, scales::percent(probability, accuracy = 1), "")),
    position = position_stack(vjust = 0.5),
    size = 4
  ) +

  # Format Y-axis as percentages and remove gap below 0
  scale_y_continuous(labels = scales::percent_format(), expand = expansion(mult = c(0, 0.05))) +

  # Set custom colors and legend labels
  scale_fill_manual(
    values = c("Negative review" = "#BDBDBD", "Positive review" = "#FFFFFF"),
    labels = c("Negative review" = "probability of posting a negative review (1/2/3 out of 5)",
              "Positive review" = "probability of posting a positive review (4/5 out of 5)")
  ) +

  # Add titles and remove axis labels where unnecessary
  labs(
    # title = "Figure 3. Probability of Posting a Negative vs. Positive Review",
    x = "Satisfaction score",
    y = NULL,
    fill = NULL
  ) +

  # Apply theme modifications to match the image
  theme_minimal() +
  theme(
    legend.position = "bottom",
    legend.direction = "vertical", # Stack legend items vertically
    legend.justification = "left", # Align legend to the left
    panel.grid.major.x = element_blank(), # Remove vertical grid lines
    panel.grid.minor = element_blank(),
    axis.line.x = element_line(color = "black"), # Solid black line for x-axis
    axis.ticks.x = element_line(color = "black"),
    plot.title = element_text(face = "bold", hjust = 0, margin = margin(b = 15)),
    axis.text = element_text(color = "black", size = 10)
  )

```

```

#-----
# Figure 4
#-----
fig_first_stage_data <- purrr::map_dfr(
  subgroup_vars,
  function(g) {
    review_data %>%
      dplyr::filter(.data[[g]] == 1) %>%
      dplyr::group_by(solicit) %>%
      dplyr::summarise(
        review_rate = mean(review, na.rm = TRUE),
        N = dplyr::n(),
        .groups = "drop"
      ) %>%
      dplyr::mutate(subgroup = g)
  }
) %>%
dplyr::left_join(subgroup_map, by = "subgroup") %>%
dplyr::mutate(
  solicit_status = dplyr::if_else(solicit == 1, "Solicited", "Unsolicited"),
  solicit_status = factor(solicit_status, levels = c("Solicited", "Unsolicited")),
  loyalty = factor(loyalty, levels = c("Silver", "Gold", "Platinum")),
  satisfaction_group = factor(satisfaction_group, levels = c("<=6", "8", "9", "10"))
)

ggplot2::ggplot(
  fig_first_stage_data,
  ggplot2::aes(x = satisfaction_group, y = review_rate)
) +
  ggplot2::geom_col(width = 0.7) +
  # Add scales = "free_y" right here:
  ggplot2::facet_grid(solicit_status ~ loyalty, scales = "free_y") +
  ggplot2::scale_y_continuous(
    labels = scales::percent_format()
  ) +
  ggplot2::labs(
    x = "Satisfaction score",
    y = "Observed probability of posting"
  ) +
  ggplot2::theme_minimal()
#-----
# Online Appendix C, Table OC1
#-----
fit_first_stage <- function(data) {
  estimatr::lm_robust(
    review ~ solicit,
    data = data,
    se_type = "HC1"
  )
}

first_stage_models <- c(
  list(Overall = fit_first_stage(review_data)),

```

```

purrr::map(
  subgroup_vars,
  ~ fit_first_stage(dplyr::filter(review_data, .data[.x] == 1))
) %>%
  purrr::set_names(subgroup_vars)
)

extract_first_stage <- function(model, subgroup_name) {
  broom::tidy(model) %>%
    dplyr::filter(term == "solicit") %>%
    dplyr::transmute(
      subgroup = subgroup_name,
      estimate = estimate,
      std_error = std.error,
      p_value = p.value,
      robust_f = statistic^2,
      N = stats::nobs(model)
    )
}

first_stage_table <- purrr::imap_dfr(
  first_stage_models,
  extract_first_stage
) %>%
  dplyr::left_join(subgroup_map, by = "subgroup") %>%
  dplyr::mutate(
    Tier = dplyr::if_else(
      subgroup == "Overall",
      "All customers",
      as.character(loyalty)
    ),
    Score = dplyr::if_else(
      subgroup == "Overall",
      "",
      as.character(satisfaction_group)
    ),
    `Solicit coefficient (SE)` = sprintf("%.2f (%.2f)", estimate, std_error),
    `Robust F` = sprintf("%.2f", robust_f),
    `p-value` = sprintf("%.3f", p_value)
  ) %>%
  dplyr::select(
    Tier,
    Score,
    `Solicit coefficient (SE)`,
    `Robust F`,
    `p-value`,
    N
  )

table_task(
  first_stage_table,
  caption = "\\label{tab:first_stage} First-stage estimates for review posting"
) %>%

```

```

collapse_rows(1, latex_hline="major")
#-----
# Table 2
#-----
fit_itt <- function(data, outcome) {
  estimatr::lm_robust(
    stats::as.formula(paste0(outcome, " ~ solicit")),
    data = data,
    se_type = "HC1"
  )
}

sig_stars <- function(p) {
  dplyr::case_when(
    is.na(p) ~ "",
    p < 0.01 ~ "***",
    p < 0.05 ~ "**",
    p < 0.10 ~ "*",
    TRUE ~ ""
  )
}

extract_solicit <- function(model) {
  broom::tidy(model) %>%
  dplyr::filter(term == "solicit") %>%
  dplyr::transmute(
    estimate = estimate,
    std_error = std.error,
    p_value = p.value,
    N = length(stats::residuals(model))
  )
}

fmt_est_se <- function(estimate, std_error, p_value) {
  paste0(
    sprintf("%.2f", estimate),
    sig_stars(p_value),
    " (",
    sprintf("%.2f", std_error),
    ")"
  )
}

itt_overall_results <- tibble::tibble(
  outcome = c("STspend", "LTspend"),
  outcome_label = c("Short-term future spending", "Long-term future spending")
) %>%
  dplyr::mutate(
    model = purrr::map(outcome, ~ fit_itt(review_data, .x)),
    result = purrr::map(model, extract_solicit)
  ) %>%
  tidyr::unnest(result) %>%
  dplyr::mutate(
    ITT = fmt_est_se(estimate, std_error, p_value),

```

```

) %>%
dplyr::select(
  Outcome = outcome_label,
  ITT,
  N
)

# itt_overall_results <- itt_overall_results %>%
#   add_row(Outcome = "No. of Observations",
#           ITT = as.character(unique(itt_overall_results$N))) %>%
#   dplyr::select(Outcome, ITT)

table_task(itt_overall_results, caption="ITT Effect on Future Spending on the Solicited Brand")
#-----
# Table 3
#-----
fit_itt_by_cell <- function(outcome, subgroup) {
  cell_data <- review_data %>%
    dplyr::filter(.data[[subgroup]] == 1)

  fit_itt(cell_data, outcome)
}

itt_hetero_long <- tidyr::expand_grid(
  outcome = c("STspend", "LTspend"),
  subgroup = subgroup_vars
) %>%
dplyr::mutate(
  model = purrr::map2(outcome, subgroup, fit_itt_by_cell),
  result = purrr::map(model, extract_solicit)
) %>%
tidyr::unnest(result) %>%
dplyr::left_join(subgroup_map, by = "subgroup") %>%
dplyr::mutate(
  outcome_label = dplyr::recode(
    outcome,
    STspend = "Short-term",
    LTspend = "Long-term"
  ),
  estimate_se = fmt_est_se(estimate, std_error, p_value),
  p_value_fmt = sprintf("%.3f", p_value)
) %>%
dplyr::arrange(loyalty, satisfaction_group, outcome)
# 1. Define a custom Goodness-of-Fit (GOF) map
# We only want to keep the Number of Observations and format it with a comma.
gof_mapping <- list(
  list("raw" = "nobs", "clean" = "No. of observations", "fmt" = function(x) format(x, big.mark=","))
)

# 2. Create a helper function to process a single tier
get_tier_df <- function(data, tier_name) {

  # Extract lists of models and name them so modelsummary uses them as columns

```

```

mods_st <- data %>% filter(loyalty == tier_name, outcome == "STspend") %>% pull(model)
names(mods_st) <- c("<=6", "8", "9", "10")

mods_lt <- data %>% filter(loyalty == tier_name, outcome == "LTspend") %>% pull(model)
names(mods_lt) <- c("<=6", "8", "9", "10")

# Summarize Short-Term models (No 'N' row needed here based on your image)
df_st <- modelsummary(
  mods_st,
  coef_map = c("solicit" = "Solicitation"),
  gof_map = NA,
  stars = c('*' = .1, '**' = .05, '***' = .01),
  output = "data.frame"
) %>%
  mutate(Horizon = "Short-term (0-5 months)")

# Summarize Long-Term models (Include 'N' row)
df_lt <- modelsummary(
  mods_lt,
  coef_map = c("solicit" = "Solicitation"),
  gof_map = gof_mapping,
  stars = c('*' = .1, '**' = .05, '***' = .01),
  output = "data.frame"
) %>%
  mutate(Horizon = "Long-term (6-10 months)")

# Bind them together and clean up the term names for standard errors
bind_rows(df_st, df_lt) %>%
  mutate(
    term = ifelse(statistic == "std.error", "", term) # Make SE rows blank in column 2
  ) %>%
  select(Horizon, term, `<=6`, `8`, `9`, `10`)
}

# 3. Build the final unified dataframe
df_final <- bind_rows(
  get_tier_df(itt_hetero_long, "Silver"),
  get_tier_df(itt_hetero_long, "Gold"),
  get_tier_df(itt_hetero_long, "Platinum")
)

df_final %>%
  table_task(output="latex") %>%
  add_header_above(c(" " = 2, "Satisfaction" = 4)) %>%
  pack_rows(
    index = c("Silver" = 5, "Gold" = 5, "Platinum" = 5),
    latex_align = "c"
  ) %>%
  collapse_rows(columns = 1, latex_hline = "none", valign = "top") %>%
  wrap_latex_table(note = "All estimations include a constant term that is not tabulated for confidence")

#-----
# Section 5 setup: LATE / IV estimates
#-----

```

```

fit_late <- function(data, outcome) {
  estimatr::iv_robust(
    stats::as.formula(paste0(outcome, " ~ review | solicit")),
    data = data,
    se_type = "HC1"
  )
}

extract_review_iv <- function(model) {
  broom::tidy(model) %>%
  dplyr::filter(term == "review") %>%
  dplyr::transmute(
    estimate = estimate,
    std_error = std.error,
    p_value = p.value,
    N = stats::nobs(model)
  )
}

#-----
# Table 4: Overall LATE
#-----

late_overall_models <- list(
  "Short-term (0-5 months)" = fit_late(review_data, "STspend"),
  "Long-term (6-10 months)" = fit_late(review_data, "LTspend")
)

modelsummary::modelsummary(
  late_overall_models,
  coef_map = c("review" = "Review"),
  gof_map = list(
    list(
      "raw" = "nobs",
      "clean" = "No. of observations",
      "fmt" = function(x) format(x, big.mark = ",")
    )
  ),
  stars = c("*" = .1, "**" = .05, "***" = .01)
) %>%
wrap_latex_table(
  notes = "All estimations include a constant term that is not tabulated for confidentiality reasons",
  caption = "Overall LATE of Posting a Review on Future Spending",
  label = "tab:late_overall"
)

#-----
# Table 5: Heterogeneous LATE
#-----

fit_late_by_cell <- function(outcome, subgroup) {
  cell_data <- review_data %>%
    dplyr::filter(.data[[subgroup]] == 1)

  fit_late(cell_data, outcome)
}

```

```

}

late_hetero_long <- tidyr::expand_grid(
  outcome = c("STspend", "LTspend"),
  subgroup = subgroup_vars
) %>%
  dplyr::mutate(
    model = purrr::map2(outcome, subgroup, fit_late_by_cell),
    result = purrr::map(model, extract_review_iv)
  ) %>%
  tidyr::unnest(result) %>%
  dplyr::left_join(subgroup_map, by = "subgroup") %>%
  dplyr::mutate(
    outcome_label = dplyr::recode(
      outcome,
      STspend = "Short-term",
      LTspend = "Long-term"
    ),
    estimate_se = fmt_est_se(estimate, std_error, p_value),
    p_value_fmt = sprintf("%.3f", p_value)
  ) %>%
  dplyr::arrange(loyalty, satisfaction_group, outcome)
# 1. Define GOF map: keep only number of observations
gof_mapping <- list(
  list(
    "raw" = "nobs",
    "clean" = "No. of observations",
    "fmt" = function(x) format(x, big.mark = ",")
  )
)

# 2. Helper function for one loyalty tier
get_late_tier_df <- function(data, tier_name) {

  mods_st <- data %>%
    dplyr::filter(loyalty == tier_name, outcome == "STspend") %>%
    dplyr::pull(model)

  names(mods_st) <- c("<=6", "8", "9", "10")

  mods_lt <- data %>%
    dplyr::filter(loyalty == tier_name, outcome == "LTspend") %>%
    dplyr::pull(model)

  names(mods_lt) <- c("<=6", "8", "9", "10")

  df_st <- modelsummary::modelsummary(
    mods_st,
    coef_map = c("review" = "Review"),
    gof_map = NA,
    stars = c("*" = .1, "**" = .05, "***" = .01),
    output = "data.frame"
  ) %>%

```

```

    dplyr::mutate(Horizon = "Short-term (0-5 months)")

df_lt <- modelsummary::modelsummary(
  mods_lt,
  coef_map = c("review" = "Review"),
  gof_map = gof_mapping,
  stars = c("*" = .1, "**" = .05, "***" = .01),
  output = "data.frame"
) %>%
  dplyr::mutate(Horizon = "Long-term (6-10 months)")

dplyr::bind_rows(df_st, df_lt) %>%
  dplyr::mutate(
    term = ifelse(statistic == "std.error", "", term)
  ) %>%
  dplyr::select(Horizon, term, `<=6`, `8`, `9`, `10`)
}

# 3. Build final table
df_late_final <- dplyr::bind_rows(
  get_late_tier_df(late_hetero_long, "Silver"),
  get_late_tier_df(late_hetero_long, "Gold"),
  get_late_tier_df(late_hetero_long, "Platinum")
)

df_late_final %>%
  table_task() %>%
  add_header_above(c(" " = 2, "Satisfaction" = 4)) %>%
  pack_rows(
    index = c("Silver" = 5, "Gold" = 5, "Platinum" = 5),
    latex_align = "c"
  ) %>%
  collapse_rows(columns = 1, latex_hline = "none", valign = "top") %>%
  wrap_latex_table(
    note = "All estimations include a constant term that is not tabulated for confidentiality reasons",
    caption = "LATE Effects of Posting a Review on Future Spending on the Reviewed Brand",
    label = "tab:late_hetero"
  )

#-----
# Section 6 setup: Romano-Wolf MHT
#-----

hcl_t_stats <- function(y, x, inv_xx) {
  beta <- inv_xx %*% crossprod(x, y)
  resid <- as.numeric(y - x %*% beta)

  n <- nrow(x)
  k <- ncol(x)

  meat <- crossprod(x * resid, x * resid)
  vcov_hcl <- (n / (n - k)) * inv_xx %*% meat %*% inv_xx

  se <- sqrt(diag(vcov_hcl))

```

```

t_stat <- as.numeric(beta) / se
names(t_stat) <- colnames(x)

t_stat
}

romano_wolf_stepdown <- function(obs_abs, boot_abs) {
  ord <- order(obs_abs, decreasing = TRUE)
  adj_ord <- numeric(length(obs_abs))

  for (j in seq_along(ord)) {
    remaining <- ord[j:length(ord)]
    boot_max <- apply(boot_abs[, remaining, drop = FALSE], 1, max)
    adj_ord[j] <- mean(boot_max >= obs_abs[ord[j]])
  }

  adj_ord <- cummax(adj_ord)

  out <- numeric(length(obs_abs))
  out[ord] <- pmin(adj_ord, 1)
  names(out) <- names(obs_abs)

  out
}

romano_wolf_itt <- function(data, subgroups, reps = 5000, seed = 12345) {

  sol_vars <- paste0("sol", subgroups)

  data_rw <- data %>%
    dplyr::mutate(
      subgroup_count = rowSums(dplyr::pick(dplyr::all_of(subgroups)), na.rm = TRUE)
    ) %>%
    dplyr::filter(subgroup_count == 1) %>%
    dplyr::select(STspend, LTspend, solicit, dplyr::all_of(subgroups)) %>%
    tidyr::drop_na()

  data_rw <- data_rw %>%
    dplyr::mutate(
      dplyr::across(
        dplyr::all_of(subgroups),
        ~ solicit * .x,
        .names = "sol{.col}"
      )
    )

  x_full <- stats::model.matrix(
    stats::as.formula(
      paste("~ 0 +", paste(c(subgroups, sol_vars), collapse = " + "))
    ),
    data = data_rw
  )

```

```

x_null <- stats::model.matrix(
  stats::as.formula(
    paste("~ 0 +", paste(subgroups, collapse = " + "))
  ),
  data = data_rw
)

inv_full <- solve(crossprod(x_full))
inv_null <- solve(crossprod(x_null))

fit_null <- function(y) {
  beta <- inv_null %*% crossprod(x_null, y)
  fitted <- as.numeric(x_null %*% beta)
  resid <- as.numeric(y - fitted)

  list(fitted = fitted, resid = resid)
}

obs_st <- hc1_t_stats(data_rw$STspend, x_full, inv_full)[sol_vars]
obs_lt <- hc1_t_stats(data_rw$LTspend, x_full, inv_full)[sol_vars]

obs_all <- c(obs_st, obs_lt)
names(obs_all) <- c(
  paste0("ST_", subgroups),
  paste0("LT_", subgroups)
)

null_st <- fit_null(data_rw$STspend)
null_lt <- fit_null(data_rw$LTspend)

boot_abs <- matrix(
  NA_real_,
  nrow = reps,
  ncol = length(obs_all),
  dimnames = list(NULL, names(obs_all))
)

set.seed(seed)

for (b in seq_len(reps)) {
  idx <- sample.int(nrow(data_rw), nrow(data_rw), replace = TRUE)

  y_st_star <- null_st$fitted + null_st$resid[idx]
  y_lt_star <- null_lt$fitted + null_lt$resid[idx]

  boot_abs[b, ] <- c(
    abs(hc1_t_stats(y_st_star, x_full, inv_full)[sol_vars]),
    abs(hc1_t_stats(y_lt_star, x_full, inv_full)[sol_vars])
  )
}

adjusted <- romano_wolf_stepdown(abs(obs_all), boot_abs)

```

```

tibble::tibble(
  subgroup = subgroups,
  `ST RW p` = adjusted[paste0("ST_", subgroups)],
  `LT RW p` = adjusted[paste0("LT_", subgroups)]
)
}

fmt_p_paren <- function(x) {
  ifelse(is.na(x), "", paste0("(", sprintf("%.3f", x), ")"))
}

#-----
# Table 6: Raw and Romano-Wolf adjusted p-values
#-----

rw_reps <- 5000L
rw_seed <- 12345L

raw_pvals <- itt_hetero_long %>%
  dplyr::select(subgroup, outcome, p_value) %>%
  dplyr::mutate(
    p_type = dplyr::recode(
      outcome,
      STspend = "ST raw p",
      LTspend = "LT raw p"
    )
  ) %>%
  dplyr::select(subgroup, p_type, p_value) %>%
  tidyr::pivot_wider(
    names_from = p_type,
    values_from = p_value
  )

rwolf_pvals <- romano_wolf_itt(
  data = review_data,
  subgroups = subgroup_vars,
  reps = rw_reps,
  seed = rw_seed
)

romano_wolf_long <- raw_pvals %>%
  dplyr::left_join(rwolf_pvals, by = "subgroup") %>%
  dplyr::left_join(subgroup_map, by = "subgroup") %>%
  dplyr::arrange(loyalty, satisfaction_group)
#-----
# Format Table 6 in the same style as Tables 3 and 5
#-----

get_rw_tier_df <- function(data, tier_name) {

  tier_data <- data %>%
    dplyr::filter(loyalty == tier_name)

  get_cell <- function(score, col) {

```

```

    tier_data %>%
      dplyr::filter(as.character(satisfaction_group) == score) %>%
      dplyr::pull(.data[[col]]) %>%
      dplyr::first() %>%
      fmt_p_paren()
  }

  tibble::tibble(
    Horizon = c(
      "Short-term (0-5 months)",
      "Short-term (0-5 months)",
      "Long-term (6-10 months)",
      "Long-term (6-10 months)"
    ),
    term = c(
      "Raw p-value",
      "Romano-Wolf p-value",
      "Raw p-value",
      "Romano-Wolf p-value"
    ),
    `<=6` = c(
      get_cell("<=6", "ST raw p"),
      get_cell("<=6", "ST RW p"),
      get_cell("<=6", "LT raw p"),
      get_cell("<=6", "LT RW p")
    ),
    `8` = c(
      get_cell("8", "ST raw p"),
      get_cell("8", "ST RW p"),
      get_cell("8", "LT raw p"),
      get_cell("8", "LT RW p")
    ),
    `9` = c(
      get_cell("9", "ST raw p"),
      get_cell("9", "ST RW p"),
      get_cell("9", "LT raw p"),
      get_cell("9", "LT RW p")
    ),
    `10` = c(
      get_cell("10", "ST raw p"),
      get_cell("10", "ST RW p"),
      get_cell("10", "LT raw p"),
      get_cell("10", "LT RW p")
    )
  )
}

df_rw_final <- dplyr::bind_rows(
  get_rw_tier_df(romano_wolf_long, "Silver"),
  get_rw_tier_df(romano_wolf_long, "Gold"),
  get_rw_tier_df(romano_wolf_long, "Platinum")
)

```

```

df_rw_final %>%
  table_task(output = "latex") %>%
  add_header_above(c(" " = 2, "Satisfaction" = 4)) %>%
  pack_rows(
    index = c("Silver" = 4, "Gold" = 4, "Platinum" = 4),
    latex_align = "c"
  ) %>%
  collapse_rows(columns = 1, latex_hline = "none", valign = "top") %>%
  wrap_latex_table(
    note = paste0(
      "Raw p-values are based on robust standard errors. ",
      "Romano--Wolf adjusted p-values are computed with ",
      rw_reps,
      " bootstrap resamples and seed ",
      rw_seed,
      ". By design, these adjusted p-values apply to the heterogeneous ITT tests and to the corresponding",
      "$~{***} p < 0.01; ~{**} p < 0.05; ~{*} p < 0.1$."
    ),
    caption = "Adjusted p-values Under Multiple-Hypothesis Testing",
    label = "tab:romano_wolf"
  )
#-----
# Section 7: Management response rates
#-----

response_rate_long <- purrr::map_dfr(
  subgroup_vars,
  function(g) {
    review_data %>%
      dplyr::filter(.data[[g]] == 1) %>%
      # Figure is about responses to posted reviews.
      # If non-reviewers have response = NA, this filter gives the same object as mean(response, na.rm = TRUE)
      dplyr::filter(review == 1 | !is.na(response)) %>%
      dplyr::summarise(
        response_rate = mean(response, na.rm = TRUE),
        n_reviews = dplyr::n(),
        .groups = "drop"
      ) %>%
      dplyr::mutate(subgroup = g)
  }
) %>%
dplyr::left_join(subgroup_map, by = "subgroup") %>%
dplyr::mutate(
  loyalty = factor(loyalty, levels = c("Silver", "Gold", "Platinum")),
  satisfaction_group = factor(
    satisfaction_group,
    levels = c("<=6", "8", "9", "10")
  )
) %>%
dplyr::arrange(loyalty, satisfaction_group)
#-----
# Values underlying Figure 5 / replication Figure 6
#-----

```

```

response_rate_table <- response_rate_long %>%
  dplyr::mutate(
    `Response rate` = scales::percent(response_rate, accuracy = 0.1)
  ) %>%
  dplyr::select(
    Loyalty = loyalty,
    Satisfaction = satisfaction_group,
    `No. of reviews` = n_reviews,
    `Response rate`
  )

response_rate_table %>%
  table_task(output = "latex") %>%
  wrap_latex_table(
    caption = "Management Response Rates by Loyalty and Satisfaction",
    label = "tab:response_rate_values"
  )
#-----
# Paper Figure 5 / replication Figure 6
#-----

fig5 <- ggplot2::ggplot(
  response_rate_long,
  ggplot2::aes(
    x = satisfaction_group,
    y = response_rate,
    group = loyalty,
    linetype = loyalty,
    shape = loyalty
  )
) +
  ggplot2::geom_line(linewidth = 0.9) +
  ggplot2::geom_point(size = 2) +
  ggplot2::scale_y_continuous(
    labels = scales::percent_format(accuracy = 1)
  ) +
  ggplot2::labs(
    x = "Satisfaction subgroup",
    y = "Share of reviews with a company response",
    linetype = NULL,
    shape = NULL
  ) +
  ggplot2::theme_minimal()

wrap_latex_figure(fig5, caption = "Management response rate by loyalty and satisfaction", label = "fig:")
# 1. Get all chunk labels in the document in order
all_chunks <- knitr::all_labels()

# 2. Find the numerical index of the 'sample-attrition' chunk
start_pos <- match("data-load", all_chunks)

# 3. Subset the list to only include chunks that come AFTER that position
appendix_chunks <- all_chunks[(start_pos + 1):length(all_chunks)]

```